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Estimating net primary production of boreal forests in Finland and Sweden from field data and remote sensing

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Abstract. We calculated annual mean stem volume increment (AMSVI) and total litter fall to produce forest net primary production (NPP) maps at 1-km^2 and half-degree resolutions in Finland and Sweden. We used a multi-scale methodology to link field inventory data reported at plot and forestry district levels through a remotely sensed total plant biomass map derived from 1-km^2 AVHRR image. Total litter fall was estimated as function of elevation and latitude. Leaf litter fall, a surrogate for fine root production, was estimated from total litter fall by forest type. The gridded NPP estimates agreed well with previously reported NPP values, based on point measurements. Regional NPP increases from northeast to southwest. It is positively related to annual mean temperature and annual mean total precipitation (strongly correlated with temperature) and is negatively related to elevation at broad scale. Total NPP (TNPP) values for representative cells selected based on three criteria were highly correlated with simulated values from a process-based model (CEVSA) at $0.5° \times 0.5°$ resolution.

At 1-km^2 resolution, mean above-ground NPP in the region was 408 g/m^2/yr ranging from 172 to 1091 (standard deviation (SD) = 134). Mean TNPP was 563 (252 to 1426, SD = 176). Ranges and SD were reduced while the mean values of the estimated NPP stayed almost constant as cell size increased from 1-km^2 to $0.5° \times 0.5°$, as expected. Nordic boreal forests seem to have lower productivity among the world boreal forests.

Keywords: Above-ground production; Annual mean stem volume increment; Grid cell; Increment; Litter fall; NPP.

Abbreviations: ABIO = Above-ground biomass; AMSVI = Annual mean stem volume increment; ANPP = Above-ground net primary production; AVHRR = Advanced Very High Resolution Radiometer; NPP = Net primary production; RS = Remote sensing; SBIO = Stem biomass; TNPP = Total net primary production; TPB = Total plant biomass.

Introduction

Changes in the global carbon budget as a result of climate change have been the subject of great concern and debate (Keeling 1973; Woodwell et al. 1978; Brown & Lugo 1980; Houghton et al. 1983; Adams et al. 1990; Tans et al. 1990; Conway et al. 1994; Keeling et al. 1996). Therefore, quantifying net primary production (NPP) for large areas is being given high priority (Bolin 1998; Anon. 1998). Currently forests cover about 40% of the ice-free land surface of the Earth (52.4×106 km^2, Waring & Running) and boreal forests account for ca. 21% of that (Gower et al. 2001). Boreal forests are of particular interest because, among all biomes, they may undergo the greatest climatically induced changes in the 21st century (Bonan et al. 1992; Myneni et al. 1997).

Currently, most field estimates of forest NPP are for small plots and are not organized in the formats needed to assess the global carbon cycle. A combination of remote sensing (RS) techniques and numerous field observations in ecological studies and routine natural resource inventories makes it possible to extrapolate plot measurements and inventory data reported at coarse resolution to two dimensional surface maps over entire landscapes for spatial pattern analyses (Tieszen et al. 1997; Brown et al. 1999; Jiang et al. 1999).

There is an urgent need to reorganize field NPP measurements to a scale suitable for analysis with climate and other data often with coarser spatial resolution. A $0.5° \times 0.5°$ cell size (Lurin et al. 1994) is widely used for global NPP studies. Point data cannot represent the NPP of the entire cell and combinations of conditions represented by driving variables that were used for the coarse-resolution model simulations may never exist anywhere in the cell. Such inconsistencies caused by differences in scales may lead to incorrect conclusions when ecophysiological models forced with coarse resolution data are used to estimate NPP, especially in mountainous or other heterogeneous areas.

Several studies have demonstrated that forest biomass can be estimated from field inventories and remotely sensed satellite data (Brown et al. 1997, 1999; Hame et al. 1997; Schroeder et al. 1997), but few studies have reported on how to estimate forest NPP from biomass over landscapes.

For regional NPP studies, a multi-scale approach incorporating both RS and field data at various scales (e.g. plot, forestry districts) is desirable because the factors controlling NPP vary with scales. For example, climate control is dominant at the coarse scale while soil and topographic effects are important at the local scale. Also, remotely sensed data with various spatial resolutions allow us to examine spatial patterns of ecosystem properties (e.g. NPP) over large areas by linking field measurements that are usually limited in numbers. This approach has been successfully applied in land use pattern analyses (i.e. the relationship between land use and the biophysical and socio-economic conditions) (Veldkamp & Fresco 1997; Walsh et al. 1997; Verburg et al. 1999).

Our study is focused on a conifer-dominated region in Finland and Sweden where field surveys and total plant biomass estimates obtained from AVHRR are available (Hame et al. 1997). In the study area, forestland occupies approximately 480 000 km^2, or 61% of the total land surface (Anon. 1997, 1999). The study area extends from ca. 55° to 70° N and 11° to 32° E. Elevations vary from sea level to 1936 m with an average of ca. 260 m. The main tree species are *Picea abies*, *Pinus sylvestris* and *Betula pubescens* and *B. pendula*, which are typical European boreal forest species.

Our general objective is to provide gridded NPP maps in Finland and Sweden based on field measurements and forest inventory data and to make comparisons with modeled NPP and climatic factors at regional scale.

Methods

We used a multi-scale methodology to link field measurements (mostly point data) with inventory data reported at forestry district level and a regional plant biomass map created using 1-km^2 AVHRR to estimate annual mean stem biomass increment (AMSBI).

The forest inventory data at plot level were used for establishing allometric relationships among different components of above-ground biomass. Other NPP components such as annual litter fall production and belowground production were either predicted from statistical models or obtained from existing literature. The detailed steps are described below. They may be summarized as follows: (1) calculate above-ground biomass (ABIO, kg.ha^{-1}) from the 1-km^2 AVHRR plant biomass map in Finland and Sweden based on literature reports; (2) convert above-ground biomass to stem biomass (SBIO, kg/ha) based on ground measurements; (3) estimate annual mean stem volume increment (AMSVI, m^3.ha^{-1}.yr^{-1}) from SBIO using statistic models developed from plot measurements; (4) adjust the AMSVI map using a ratio map and the district-level mean values of AMSVI calculated from regression equations (see below); (5) validate the modified AMSVI map using independent district-level AMSVI data that were not used for developing equation 2; (6) estimate annual total litter fall from latitude and elevation (Lonsdale 1988) and leaf litter fall from total litter fall by forest type; (7) produce ANPP and TNPP based on commonly used partitioning values between ANPP and below-ground NPP; and (8) aggregate the 1-km^2 NPP maps to 0.5° × 0.5° grid cells using a GIS focal mean function.

Total plant biomass map

A TPB map for Finland and Sweden had been produced by Hame et al. (1997) (Box 1, Fig. 1). For the map, the TPB had been first estimated in small areas using ground measurements of biomass and high-resolution Landsat TM data from Finland. The relationship was applied to the larger area using AVHRR image mosaic after intensity values of the original spectral channels between the two different sensors were inter-calibrated. Research literature suggested that the model underestimates TPB by as much as 20% (Kauppi et al. 1995). However, the underestimation may not have been seriously harmful in the context of this paper because we are interested in NPP and not in the absolute biomass. Also, our satellite-driven estimates were calibrated using a much larger set of ground observations. Therefore, smaller errors were expected in our NPP estimates than in the original TPB estimates.

Converting to cell-level AMSVI values

First, we used a factor of 0.8 to convert the TPB to above-ground biomass (ABIO Mg.ha^{-1}) based on literature reports (Box 2, Fig. 1). For example, Cairns et al. (1997) reported that above-ground biomass accounted for about 82% of total below- and above-ground biomass at boreal latitudes. Other studies indicate that root:shoot ratio is about 1:3 (e.g. 75% was partitioned to above-ground biomass) for coniferous forests (Cannell 1982; Korner 1994).

Second, we estimated SBIO from ABIO based on 660 NFI temporary and permanent plots sampled over a 5-yr period (1994-1998) which we obtained from the Swedish University of Agricultural Sciences (Kempe pers. comm.). Details on permanent plot measurements

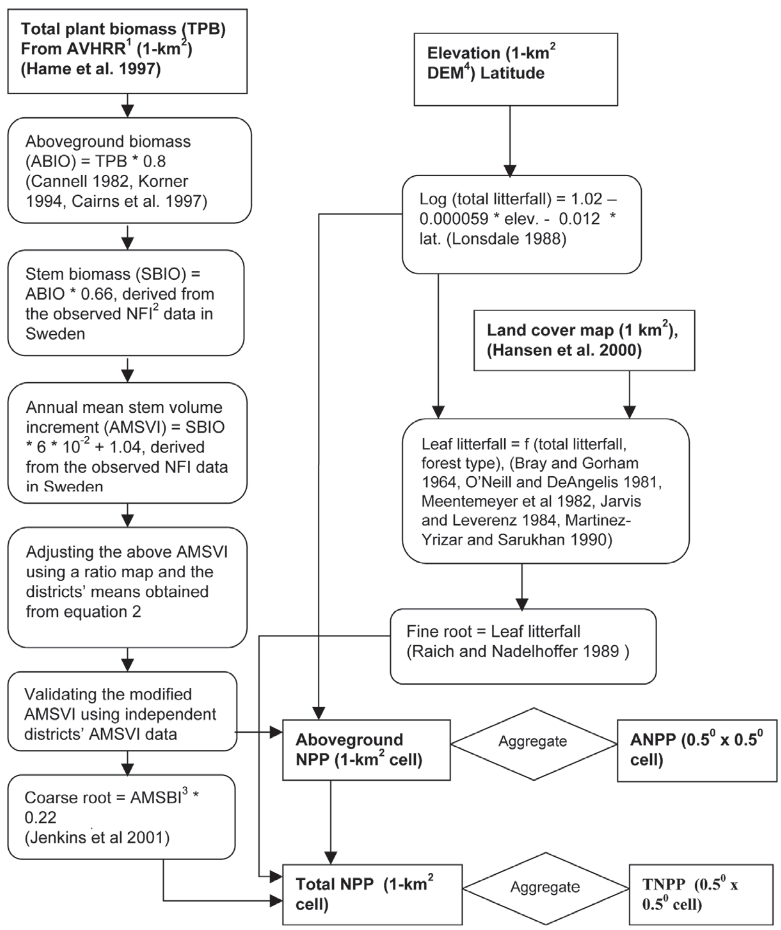

Fig. 1. Flow chart for predicting above-ground and total net primary production (NPP) maps at 1-km^2 and 0.5° × 0.5° cell sizes in Finland and Sweden. [1] = Advanced Very High Resolution Radiometer; [2] = National Forest Inventory; [3]AMSBI was converted from AMSVI using equation 3; [4] = Digital Elevation Models. Inputs and outputs are in bold type and rectangular boxes; intermediate steps are in boxes with rounded corners.

are described by Söderberg (1997). All plots used in this study are located in three representative 0.5° cells distributed across the study area along a NE-SW transect, centred at 65.15° N and 20.15° E, 61.15° N and 15.15 ° E, and 57.15° N and 13.15° E, respectively. For each plot, we acquired data on latitude, longitude, AMSVI, SBIO, branch biomass (kg/ha), and needle biomass (kg/ha). The plots contained representative coniferous, mixed, and deciduous forests, with a majority of coniferous forests (82%). The forest type for a given plot was determined from a larger plot (20 m radius) having the same plot centre as the smaller plot. The data suggested that, on average, SBIO amounted to 66% of ABIO (Box 3, Fig. 1).

Third, we derived a statistic model using the NFI plot data in Sweden to estimate AMSVI (Box 4, Fig. 1):

$$\text{AMSVI} = \text{SBIO} * 6 * 10^{-2} + 1.04;\ r^2 = 0.55\ (p < 0.001) \qquad (1)$$

The productivity probably should asymptote or decrease after reaching a certain biomass level, but the available data did not support computation of such a model.

Modifying AMSVI using district means and a ratio map

We created a map of relative AMSVI within forestry districts by dividing the AMSVI value for each cell (calculated as described above) by the mean of the cell values for that district. We then multiplied the relative values by absolute values of mean AMSVI by district estimated from NFI data for 31 districts in Sweden (Kempe pers. comm.) and 14 in Finland (Anon. 1999); more details are given below. This ratio map preserved the spatial variation of AMSVI we could obtain from the remote-sensing data, while improving accuracy by using empirical estimates of AMSVI based on a large sampling size over a long period. For example, the design for Swedish NFI is a systematic cluster sampling with partial replacement of plots. In total, there are about 30 000 permanent plots distributed on about 4300 clusters. At present the permanent plots are measured at an average interval of 7 - 8 yr, which means that about 430 clusters or 3000 permanent plots are measured every year. In addition about 9000 temporal plots distributed on 1000 clusters are measured every year (Anon. 1997)

Our NFI-based estimates of AMSVI by district were generated by regressing reported district AMSVI on location. For each of the districts, latitudes and longitudes at the geometric centres were identified. These districts were systematically divided into two groups (e.g. odd numbers vs. even numbers) except districts 28 and 29 in Sweden where only combined mean AMSVI value was available. One group was used for establishing the following statistic model (Eq. 2) between the district AMSVI and geographic location. The district AMSVI values in another group were reserved for model validation.

$$\mathrm{AMSVI}_{\mathrm{district}} = 42.2 + 0.14{*}\mathrm{Longitude} - 0.653{*}\mathrm{Latitude} \quad (2)$$
$$r^2 = 0.85, p < 0.001$$

where $\mathrm{AMSVI}_{\mathrm{district}}$ ($m^3.ha^{-1}$) is the estimated mean value for a given district after incorporating climatic effects (because in the region annual mean temperature and annual mean total precipitation are well correlated with latitude and longitude). Eq. 2 was applied to all districts to generate a district-level AMSVI map for the entire region. The map was then multiplied by the ratio map to produce the modified AMSVI map at 1-km^2 resolution (Box 5, Fig. 1).

Before the 1-km^2 modified AMSVI map was used to generate NPP maps, we aggregated to district level for comparison with the reported district data that were not used in developing Eq. 2 (Box 6, Fig. 1).

Finally the modified AMSVI ($m^3.ha^{-1}.yr^{-1}$) map was converted to AMSBI, $kg.ha^{-1}.yr^{-1}$) using Eq. 3, developed by Hame et al. (1997) based on published literature.

$$\mathrm{AMSBI} = 576\ (kg.m^{-3}) * \mathrm{AMSVI}\ \ (m^3.ha^{-1}.yr^{-1}) \qquad (3)$$

Estimating total and leaf litter fall from elevation, latitude, and forest types

We estimated annual total litter fall production using the following equation developed by Lonsdale (1988) from 181 forest sites around the world:

$$Y = 10^{\,(1.02 - 0.000059X_1 - 0.012X_2)} \qquad r^2 = 0.63 \qquad (4)$$

($Y = Mg.ha^{-1}.yr^{-1}$; X_1 = elevation in m; X_2 = latitude in decimal degrees). The 1-km^2 gridded elevation data were downloaded from the NOAA’s National Geophysical Data Centre (http://www.ngdc.noaa.gov/seg/topo/globe.shtml). Total litter fall includes all organic debris deposited on the soil and may include twigs, bark, fruits, flowers, nuts, bud scales, and insect bodies, as well as leaves.

Among various multiple regression models based on latitude, altitude, and precipitation, which were evaluated, eq. 4 was the best one at the global scale (Lonsdale 1988). Other studies suggested that latitude is a powerful variable in predicting total litter fall worldwide (Bray & Gorham 1964; Meentemeyer 1982; Vogt et al. 1986), although other factors such as species stand age, and soil nutrient and moisture characteristics (Bray & Gorham 1964) can also contribute to litter fall production at the local scale.

Fine root production is another component needed for calculating total NPP, but it is difficult to measure at any scale. We used leaf litter fall as a surrogate, as suggested by Raich & Nadelhoffer (1989). We calculated leaf litter fall from total litter fall based on forest cover types mapped at 1-km^2 resolution (Hansen et al. 2000). Several studies reported that the average proportion of leaf in total litter fall was about 70% (Bray & Gorham 1964; Meentemeyer 1982; Jarvis & Leverenz 1984; Martinez-Yrizar & Sarulhan 1990). It was concluded from O’Neill & DeAngelis (1981) that, in general, wood accounted for a higher proportion of total litter in coniferous forests than in angiosperm forests. Therefore, we estimated leaf litter fall from total litter fall using 65% for coniferous forests, 75% for deciduous forests, and 70% for mixed forest.

The four forest types in the study area were, according to Hansen et al. (2000), (1) evergreen needle-leaf; (2) deciduous broad-leaf; (3) mixed forest; (4) woodland (tree canopy cover > 40% and < 60%). Ca. 48% of forested land in the study area was classified as woodland where lower leaf litter fall production was expected, owing to the reduced canopy cover. Thus, leaf litter fall per unit area calculated from equation 4 was reduced by a factor of 0.5 for woodland.

Mapping above-ground and total NPP at 1-km^2 and 0.5° × 0.5° cell sizes

To estimate TNPP, we assumed that coarse root increment is about 22% of AMSBI (D.J. Barrett subm.; Jenkins et al. 2001). Consequently, ANPP was calculated as the sum of AMSBI and annual total litter fall production while TNPP was calculated as the sum of ANPP, fine root production and coarse root production.

AMSBI, total litter fall production, ANPP, and TNPP were calculated at 1-km resolution. These 1-km^2 NPP maps were aggregated to 0.5° × 0.5° using a focal-mean function in GIS. Any 1-km^2 cells within the window with no data were excluded during the calculation.

Other validation

No observed NPP data were available at 1-km^2 or half-degree cell size to test the results. However, total litter fall reported from literature (Bray & Gorham 1964), leaf litter fall measurements from an independent study in the region (Kouki & Hokkanen 1992), and published AMSVI data in Sweden and Finland were available for comparisons with the estimated values. We also compared our TNPP estimates for the selected 0.5° × 0.5° cells with mean TNPP values over a 15-yr period (1981-1995) estimated from the CEVSA (Carbon Exchange between Vegetation, Soil, and the Atmosphere), a process-based biogeochemical ecosystem model. The model describes the dynamic changes in terrestrial NPP, carbon storage in vegetation mass and soil, and carbon sequestration (net ecosystem production) (Cao & Woodward 1998).

We further evaluated our NPP estimates by correlating the 0.5° × 0.5° TNPP values with major environmental variables (e.g., annual mean temperature, annual mean total precipitation, and elevation) at the same resolution (0.5° × 0.5°). The climatic data were derived from monthly observations from 1961-1990 (New et al. 2000). The elevation data were aggregated from National Geophysical Data Centre's 1-km^2 products.

Results

Production for 1-km^2 cells

Spatial AMSBI patterns showed a strong gradient from North (63 g.m^{-2}.yr^{-1}) to South (438 g.m^{-2}.yr^{-1}) across the region. The average AMSBI over the entire region was 230 g.m^{-2}.yr^{-1} with SD of 121 g.m^{-2}.yr^{-1}. About 70% of forest area had AMSBI values ranging from 98 to 351 g.m^{-2}.yr^{-1} at 1-km^2 resolution.

Total litter fall production varied from 141 g.m^{-2}.yr^{-1} in northern parts of the region to 227 g.m^{-2}.yr^{-1} in southern parts, with an average of 180 g.m^{-2}.yr^{-1} in the study area. Leaf litter fall production followed a similar spatial pattern and varied from 70 g.m^{-2}.yr^{-1} in the north to 167 g.m^{-2}.yr^{-1} in the south, with an average of 104 g.m^{-2}.yr^{-1} in the study area. The trend in spatial patterns

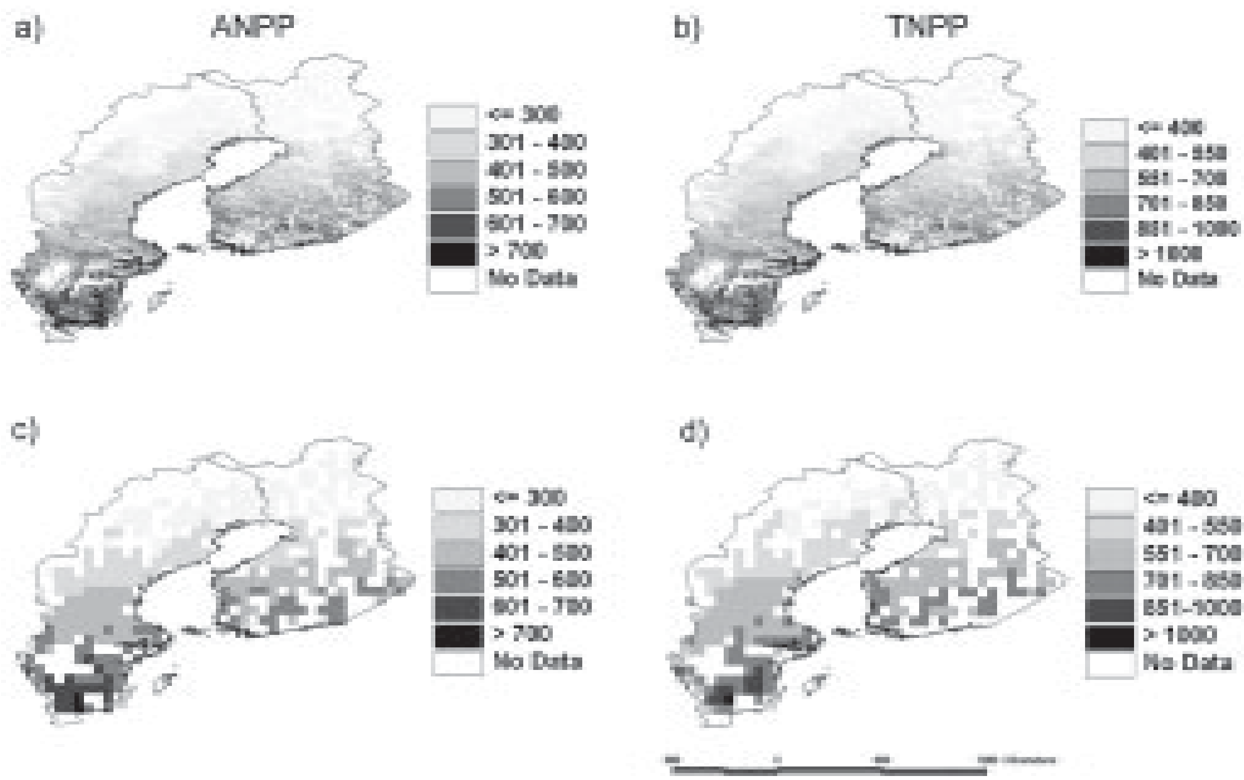

Fig. 2. Spatial patterns of ANPP (AMSBI + Total Litter fall) and TNPP (ANPP + Roots) in unit of g.m^{-2}.yr^{-1} in Finland and Sweden: **a.** 1-km^2 ANPP; **b.** 1-km^2 TNPP; **c.** 0.5° × 0.5° ANPP; **d.** 0.5° × 0.5° TNPP. NPP gradient in southeast-northwest direction was clearly observed, especially at coarser resolution.

of litter fall production was similar to that of AMSBI in general. Frequency analysis indicated that 73.4% of forest land had total litter fall production ranging from 160 to 200 g.m^{-2}.yr^{-1}.

We predicted that ANPP ranged from 172 to 1091 g.m^{-2}.yr^{-1} (mean 408, SD 134) in Finland and Sweden (Fig. 2a). Total NPP varied from 252 to 1426 g.m^{-2}.yr^{-1} (mean 563, SD 176) (Fig. 2b). Ca. 95% of forest land values of ANPP and TNPP fell within 2 SD of the mean.

Production at half degree resolution

Aggregation to 0.5° × 0.5° reduced the range and SD, but the means stayed almost constant. For example, ANPP estimates at 0.5° × 0.5° ranged from 204 to 752 g.m^{-2}.yr^{-1} with a mean value of 407 g.m^{-2}.yr^{-1} (SD 119 g.m^{-2}.yr^{-1}) (Fig. 2c); TNPP estimates at 0.5° × 0.5° ranged from 305 to 1005 g.m^{-2}.yr^{-1} with a mean value of 578 g.m^{-2}.yr^{-1} (SD 154 g.m^{-2}.yr^{-1}) (Fig. 2d).

We selected 82 0.5° × 0.5° cells in the region that met the following three criteria: (1) they were within the coniferous forest zone (< 66° N); (2) the forest cover exceeded 80%; (3) evergreen needle-leaf forest exceeded 50% area of each 0.5° cell. Thus, these cells may provide the representative estimates of ANPP and TNPP for fully stocked coniferous forests of the region for global NPP modelling and validation purposes. The 82 cells have been incorporated into the International Geosphere-Biosphere Programme (IGBP) Data Information System (DIS) global primary production data initiative (GPPDI) data set (Zheng et al. 2003).

Validation of modelled outputs

The estimated total litter fall productions at 1-km^2 resolution (141-227 g.m^{-2}.yr^{-1}) compared reasonably well with the published data (160-260 g.m^{-2}.yr^{-1}) based on point measurements in the region (Bray & Gorham 1964). Another study presented a mean leaf litter fall production of 104 g.m^{-2}.yr^{-1} (varying from 18 to 213 g.m^{-2}.yr^{-1}) for a Scots pine (*Pinus sylvestris*) stand in southern Finland over a 24-yr period (Kouki & Hokkanen 1992). Our estimated leaf litter fall for that location was 122 g.m^{-2}.yr^{-1}.

Our estimated AMSBI values at the district level (aggregated from 1-km^2 resolution estimates) across the region are correlated well (r^2 = 0.66) with the observed data in the reserved group that was not used for model development (Fig. 3). Taking the region as a whole, the estimated mean AMSBI was 230, compared to the observed value of 236 g.m^{-2}.yr^{-1}.

Our NPP estimates were significantly correlated with major environmental factors at 0.5° × 0.5° resolution (Fig. 4). The lower *r*-value (0.37) was still significant at

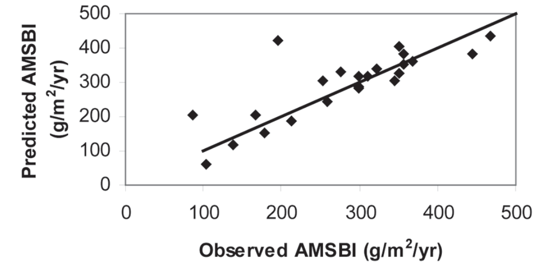

Fig. 3. Relationship between the predicted annual mean stem biomass increment (AMSBI) and observed district values that were not used in model development across Finland and Sweden. Each point represents a forestry district (r^2 = 0.66, N = 23).

$p < 0.001$ due to our large sample size (N = 261).

We compared our 0.5° × 0.5° TNPP values with the values estimated from CEVSA (Cao & Woodward 1998), a process-based terrestrial ecosystem model. A strong relationship between the two estimates (Fig. 5) is observed (r^2 = 0.82). Despite the high correlation, the model generated about 30% higher estimates than our field estimated values at the low end of TNPP values. If we are to understand global patterns of NPP, it will be important to examine such discrepancies in model predictions.

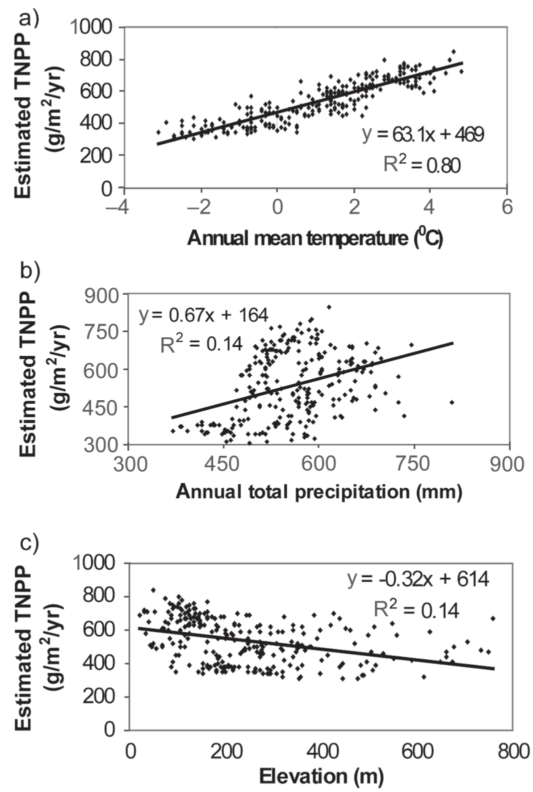

Fig. 4. Correlations between the estimated 0.5° × 0.5° TNPP estimates and environmental factors in the region (n = 261): **a.** Positive relation with annual mean temperature (1961-1990); **b.** Positive relation with annual mean total precipitation (1961-1990); **c.** Negative relation with elevation.

Discussion

Our discussion focuses on three topics: results and what determines pattern, possible errors in NPP estimates, and potential use of the methodology. Published TNPP values for boreal forests range from 109 to 1827 g.m^{-2}.yr^{-1} with an average of 892 g.m^{-2}.yr^{-1} worldwide (Gower et al. 2001). TNPP in Nordic countries is at the lower end of the world boreal forests (averaging 579 g.m^{-2}.yr^{-1} and ranging from 109 to 973 g.m^{-2}.yr^{-1}) (Ågren et al. 1980; Bergh 1997; Gower et al. 2001). Our TNPP estimates (mean 563 g.m^{-2}.yr^{-1} at 1-km^2 spatial resolution and 578 g.m^{-2}.yr^{-1} at 0.5° × 0.5° resolution) seemed to be quite compatible with these studies considering the fact that all the above reports were based on plot or stand level estimates while our estimates were based on gridded NPP for the entire region and were therefore likely to include all forest conditions.

The spatial pattern of NPP is correlated with climate at the broad scale and is somewhat easier to see at the 0.5° × 0.5° resolution (Fig. 2c, 2d) than those at 1-km^2 resolution (Fig. 2a, 2b). Regional forest NPP is positively correlated with temperature and precipitation and negatively with elevation (Fig. 4). This evidence supports the previous reports that forest productivity was positively related to temperature and precipitation over large scales (Lieth 1975; Gower et al. 2001).

At the country level, we underestimated the mean AMSBI by about 23 g.m^{-2}.yr^{-1} (– 9.3%) for Sweden and overestimated the mean AMSVI for Finland by the same amount (10.5%), although the difference between the observed and estimated means for the region was much smaller (6 g.m^{-2}.yr^{-1}, – 2.5%). Removing one anomalous point (196, 420) raised the r^2 value (0.84) between our estimated and observed AMSBI (Fig. 3). The point represents forestry district 31 in Sweden, which is a very special island with limestone bedrock. The soils are generally very shallow, resulting in low-production pine forests compared to other districts with similar climate (Kempe pers. comm.).

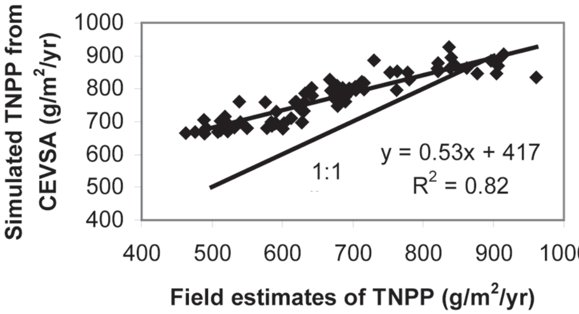

Fig. 5. Comparison between the TNPP estimated from this study and the simulated TNPP from a process-based terrestrial ecosystem model (Carbon Exchanges between Vegetation, Soil, and Atmosphere-CEVSA) for the 76 0.5° cells (six cells from the CEVSA with value of zero were excluded).

Several factors might have contributed to errors in NPP estimation. In this study, annual mean stem volume increments were estimated from biomass using linear regressions that may not apply to stands at all ages. For example, it has been reported that the ratio of NPP to total plant biomass generally declines with increase of total plant biomass in coniferous forests (Rodin & Bazilevich 1967). Also, plant biomass generally accumulates rapidly during the first 30 to 60 yr, then levels off while peak NPP values are reached between less than 20 yr to more than 60 yr in pine forest ecosystems, then decline (Cousens 1974; Nishioka 1980; Gholz & Fisher 1982; Pearson et al. 1984; Gower et al. 1994; Knight et al. 1994). In both countries, the percentage of old stands increased from south to north (e.g. from 9 to 24% in Sweden and from 12.5 to 30% in Finland), which might in part explain why our NPP estimates tended to generate higher errors in the northern part of the region than those in the southern part. For example, mean absolute percentage of differences between the estimated and observed district AMSBI values for the northern part of the region was 26%, compared to 11% for the southern part.

Additional errors in NPP estimates might also have been introduced because statistical models established using data from one area were applied to another. For instance, statistical models and allometric relations among different biomass components were derived using NFI plot data from Sweden and were applied to the entire area to estimate AMSVI (Eq. 1). Despite this less-than-ideal situation, we feel confident in our AMSVI estimates (consequently, the NPP estimates), because the multi-scale approach incorporated (1) coarser resolution district data that reflect general patterns across the study area; (2) point data and RS observations with finer resolutions that can reveal spatial variations within the sub-regions.

There is an emerging movement toward better integration of land inventories, satellite imagery, and research observations at multiple scales to examine spatial patterns of ecosystem properties and processes over landscape level or large. Remote sensing data can play an important role in such applications because of its advantage to monitor surface variations over large area with various spatial resolutions and temporal frequencies. The multi-scale methodology used in this study can be a practical and valuable tool for bringing together large data sets at various scales accumulated for different purposes (RS data, existing inventory data at plot, district, and county levels) to provide estimates of regional NPP. Our 0.5° × 0.5° NPP products can fill a growing need for a broad-scale NPP dataset that is suitable for comparison with estimates from various global NPP models at the same spatial resolution (Lurin et al. 1994; Cramer et al. 1999; Ruimy et al. 1999; Zheng et al. 2003). Coarse-resolution NPP (i.e. 0.5°) estimates also allow direct correlations with key controlling variables (e.g., precipitation, temperature, elevation) across broader scales.

Results from this study (1) are consistent with the previous report that Nordic boreal forests have low productivity among world boreal forests; (2) have clearly demonstrated spatial patterns of the NPP in the region; and (3) are useful for comparisons with other NPP values estimated elsewhere to examine geographical trends and spatial patterns of NPP worldwide.

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